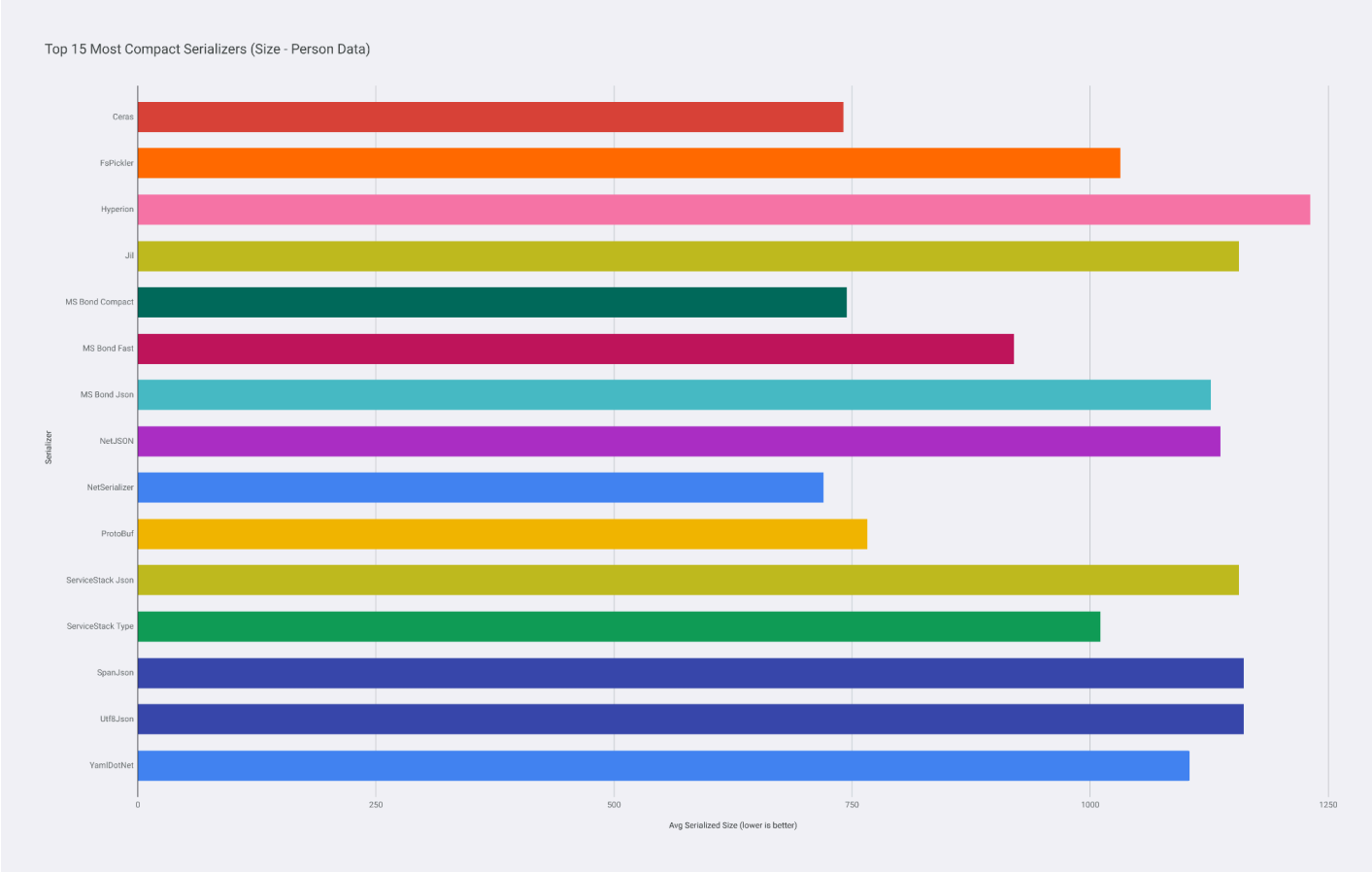
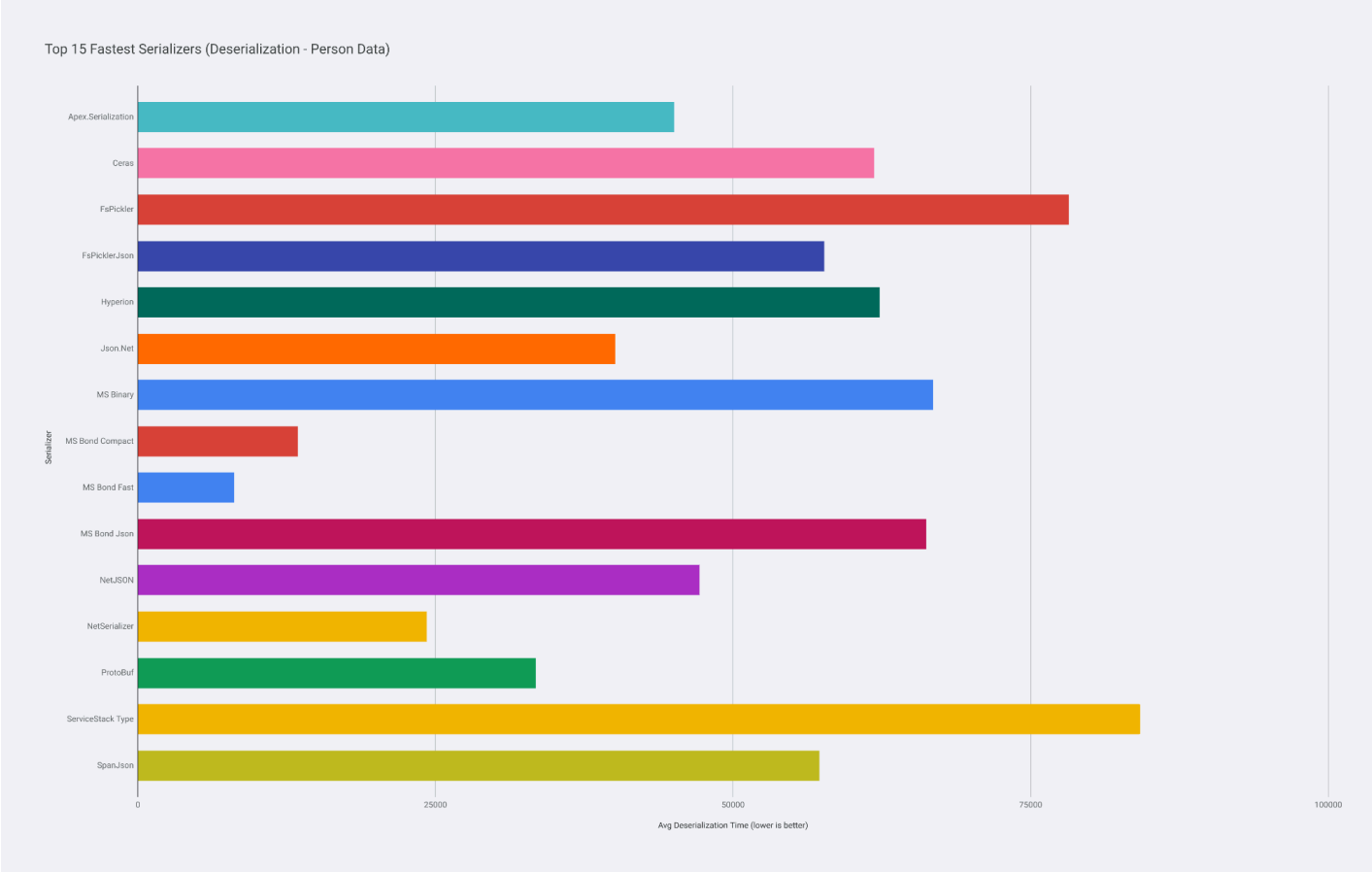
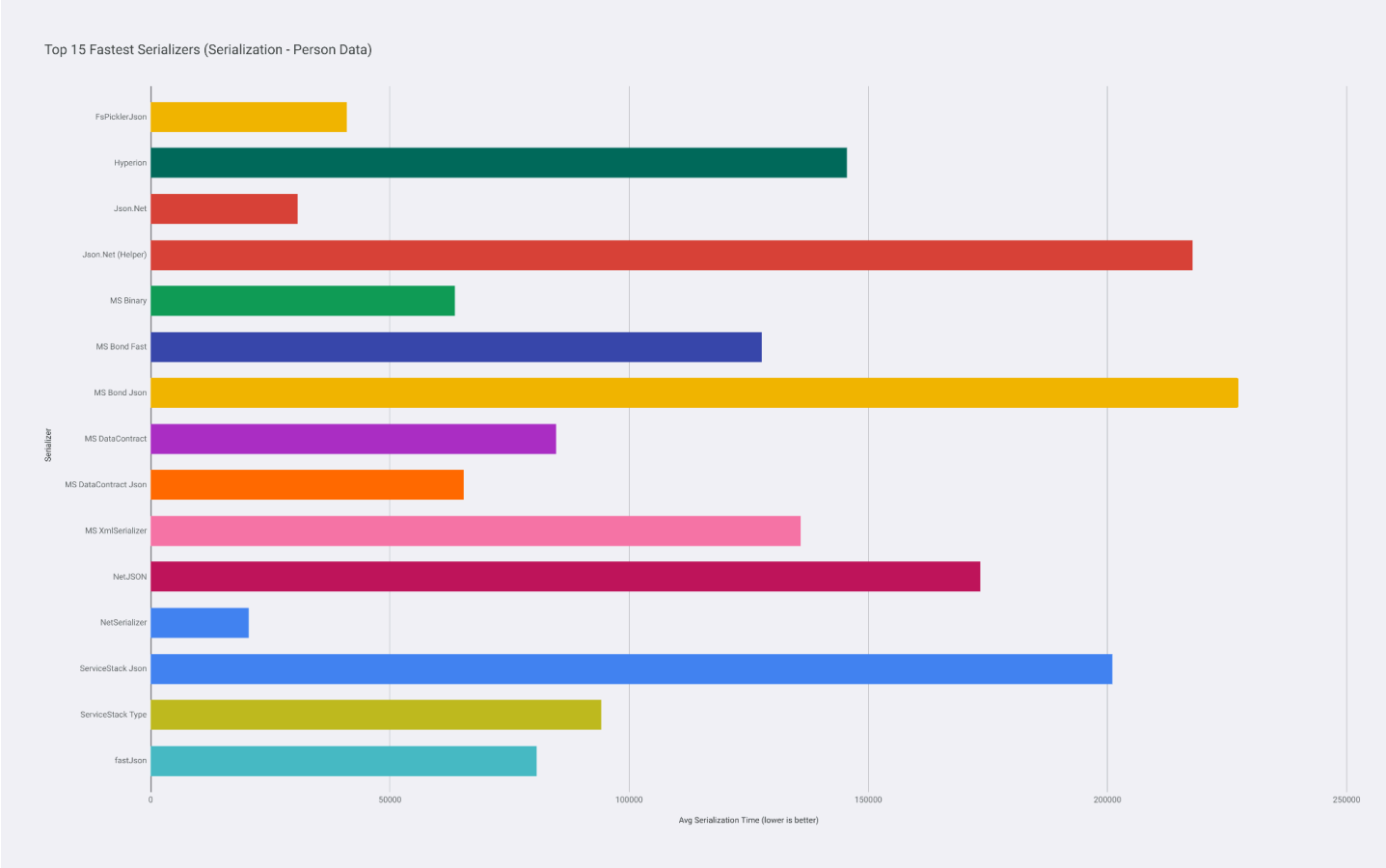
Benchmark Analysis

Analysis is created in one-click within Google Sheets with Gemini.




Top 15 Fastest Serializers (Deserialization - Person Data)
Serializer
Apex.Serialization
Ceras
FsPickler
FsPicklerJson
Hyperion
Json.Net
MS Binary
MS Bond Compact
MS Bond Fast
MS Bond Json
NetJSON
NetSerializer
ProtoBuf
ServiceStack Type
SpanJson
0
25000
50000
75000
100000
Avg Deserialization Time (lower is better)



Based on the analysis of the serializer benchmark data, here are the key insights regarding the performance and efficiency of various serialization libraries, specifically focusing on the "Person" test data as a representative sample.

Key Takeaways

- **NetSerializer is the most balanced high-performer.** It ranks as the fastest for serialization, the most compact in terms of output size, and the third-fastest for deserialization. This makes it an excellent choice for general-purpose high-performance needs.
- **MS Bond excels in Deserialization speed.** The MS Bond Fast and MS Bond Compact variants are significantly faster at deserializing data than almost any other library, which is critical for read-heavy applications.
- **Trade-offs between Speed and Size.** While Json.Net is very fast at serialization, it produces larger payloads compared to binary formats like NetSerializer or ProtoBuf. Libraries like Ceras and MS Bond Compact offer a better balance for minimizing network/storage overhead.

Performance Details (Person Test Data)

1. Serialization Speed (Fastest)

Serialization time measures how long it takes to convert an object into a storable/transferable format.

- **NetSerializer**: 20,492 avg time
- **Json.Net**: 30,733 avg time
- **FsPicklerJson**: 41,015 avg time
- **MS Binary**: 63,599 avg time
- **MS DataContract Json**: 65,482 avg time

2. Deserialization Speed (Fastest)

Deserialization time measures the speed of reconstructing an object from its serialized form.

- **MS Bond Fast**: 8,080 avg time
- **MS Bond Compact**: 13,445 avg time
- **NetSerializer**: 24,289 avg time
- **ProtoBuf**: 33,442 avg time
- **Json.Net**: 40,110 avg time

3. Payload Compactness (Smallest Size)

Smaller sizes reduce bandwidth usage and storage costs.

- **NetSerializer**: 720.0 bytes
- **Ceras**: 741.0 bytes
- **MS Bond Compact**: 744.5 bytes
- **ProtoBuf**: 766.0 bytes
- **MS Bond Fast**: 920.5 bytes

The dataset contains benchmarks for 33 different serializers across 7 test data types, including Person, Integer, Telemetry, SimpleObject, StringArray, EDI_835, and ObjectGraph. While the rankings above focus on the Person object, these trends generally hold across other data types, with binary serializers consistently outperforming text-based ones (like XML or JSON) in both speed and size.